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PS C:\Users\minhm\Desktop\Classes\Fall 2025\CS253P\Hampr\cs-253p-hw-hampr-machine-service> npm test
npm WARN config global '--global', '--local' are deprecated. Use '--location=global' instead.

> hampr-base@1.0.0 test
> jest
PASS test/api.test.ts
PASS test/simulation.test.ts
• Console

console.log



| (index) | Resource                 | Run 1 Units | Run 1 %  | Run 2 Units | Run 2 %  | Run 3 Units | Run 3 %  | Run 4 Units | Run 4 %  |
|---------|--------------------------|-------------|----------|-------------|----------|-------------|----------|-------------|----------|
| 0       | 'IdentityProviderClient' | 3840256     | '69.18%' | 3840256     | '69.38%' | 3840256     | '69.33%' | 3840256     | '69.19%' |
| 1       | 'SmartMachineClient'     | 32256       | '0.58%'  | 32256       | '0.58%'  | 32256       | '0.58%'  | 32256       | '0.58%'  |
| 2       | 'MachineStateTable'      | 1655008     | '29.82%' | 1655008     | '29.90%' | 1655008     | '29.88%' | 1655008     | '29.82%' |
| 3       | 'DataCache'              | 22422       | '0.40%'  | 22422       | '0.41%'  | 22422       | '0.40%'  | 22422       | '0.40%'  |



at Object.<anonymous> (test/simulation.test.ts:160:13)

console.log



| (index) | Run | Cache Hits | Cache Misses | Hit Rate |
|---------|-----|------------|--------------|----------|
| 0       | 1   | 3843       | 2165         | '63.96%' |
| 1       | 2   | 3901       | 2125         | '64.74%' |
| 2       | 3   | 3876       | 2114         | '64.71%' |
| 3       | 4   | 3841       | 2099         | '64.66%' |



at Object.<anonymous> (test/simulation.test.ts:161:13)

console.log



| (index) | Run | Cache Hits | DB Accesses | Hit/Access Ratio |
|---------|-----|------------|-------------|------------------|
| 0       | 1   | 3843       | 6459        | '0.5950'         |
| 1       | 2   | 3901       | 6459        | '0.6040'         |
| 2       | 3   | 3876       | 6459        | '0.6001'         |
| 3       | 4   | 3841       | 6459        | '0.5947'         |



at Object.<anonymous> (test/simulation.test.ts:162:13)

Test Suites: 2 passed, 2 total
Tests: 12 passed, 12 total
Snapshots: 0 total
Time: 2.462 s, estimated 3 s
Ran all test suites.
```

From the performance statistics, it seems that the external API IPC is being called extremely frequently and our database MachineStateTable compared to SmartMachineClient and DataCache cache. It is likely due to the lack of other implementations in the machine usage such as retrieval, delivery, timer, stop, etc., so that their usage aren't as high as the others. And just from the simple system implementations, I would say our cache hit rate aren't great and most likely due to high number of machines available per location.